

ZIHAO(MATTHEW) WANG

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Github ◇ Orcid ◇ NASA ADS

EDUCATION AND EXPERIENCE

Peking University

PhD student in Astronomy

Advisor: Prof. Kohei Inayoshi

Sept 2025 - June 2030 (expected)

Nanjing University

B.S. in Astronomy (Advanced Class)

GPA: 4.54/5 (Rank: 1/ 42)

Thesis: *Estimation and physical origin of high-redshift galaxy properties in the JWST era*

Advisor: Prof. Tao Wang

Sept 2021 - June 2025

Massachusetts Institute of Technology

Visiting research student in Physics

Advisor: Dr. Xuejian Shen & Prof. Mark Vogelsberger

June 2024 - Decem 2024

RESEARCH INTERESTS

High-redshift AGN & Little Red Dots: Theoretical modeling and predictions of the origin and evolution of Little Red Dots; Large-scale environments of high- z AGN; High- z AGN population in cosmological simulations

High-redshift galaxies & star formation: Multi-scale star formation at high- z from cloud and cluster to galactic scales; Diverse morphology of high- z galaxies; Metal enrichment

Machine learning in astrophysics: Use of machine learning techniques in galaxy parameter estimation, classification, and target hunting.

PUBLICATIONS

Publications (as of 2026/8): 4 first-author (2 peer-reviewed) and 3 contributing-author (2 peer-reviewed) papers; 89 total citations.

First-author papers

- [1] **Z. Wang**, X. Shen et al., The THESAN-ZOOM project: clumpiness of high-redshift galaxies and its connection to bursty star formation, *arXiv*, 2026 [2608.19308]
- [2] **Z. Wang**, F. Jiang et al., Halo assembly bias in the early Universe: a clustering probe of the origin of the Little Red Dots, *arXiv*, 2026 [2603.15736]
- [3] **Z. Wang**, T. Wang et al., SHAPE: I. A SOM-SED hybrid approach for efficient galaxy parameter estimation leveraging JWST, *A&A*, 2026 [2510.00187]
- [4] **Z. Wang**, X. Shen et al., The THESAN-ZOOM project: star formation efficiency from giant molecular clouds to galactic scale in high-redshift starbursts, *MNRAS*, 2025 [2505.05554]

Contributing-author papers

- [5] W. McClymont, V. Belokurov et al., including **Z. Wang**, Globular cluster abundance patterns inherited from giant molecular clouds, *arXiv*, 2026 [2607.05509]
- [6] W. McClymont, S. Tacchella et al., including **Z. Wang**, The THESAN-ZOOM project: mystery N/O more — uncovering the origin of peculiar chemical abundances and a not-so-fundamental metallicity relation at $3 < z < 12$, *MNRAS*, 2026 [2507.08787]

[7] X. Shen, R. Kannan et al., including **Z. Wang**, The THESAN-ZOOM project: star formation efficiencies in high-redshift galaxies, *MNRAS*, 2026 [2503.01949]

TALKS AND PRESENTATIONS

- [1] Conference Talk, Guoshoujing annual meeting, Haikou, China
Where Little Red Dots live tells us what they are 2026
- [2] Conference Talk, CSST annual meeting, Kunming, China
Introducing SHAPE: A SOM-SED Hybrid Approach for Efficient galaxy Parameter Estimation 2025
- [3] Conference Poster, Sino-German conference on Galaxy Formation, Chengdu, China
Elevated star formation efficiency at high-z: from a giant molecular cloud perspective 2025

HONORS AND AWARDS

Outstanding Undergraduate Thesis of Nanjing University (the special prize) 2025

Zheng-Gang Oversea Scholarship of Nanjing University 2024

Elite Program Scholarship (the special prize) 2024

People's Scholarship (the first prize) 2022, 2023, 2024

National Astronomical Observatories Scholarship 2022

SKILLS/HOBBIES (* DENOTES EXTENSIVELY)

Programming Languages	Python*, C/C++, HTML
Machine Learning Tools	SOM*, CNN, Random-Forest
Language	Mandarin (Native), English (Proficient)
Simulations	<i>Thesan-Zoom*</i> , <i>LUMINA*</i> , <i>ShinUchuu</i> , <i>IllustrisTNG</i> , <i>FIRE</i>
Hobbies	cinema and photographing

PROFESSIONAL REFERENCES

Prof. Kohei Inayoshi
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Prof. Mark Vogelsberger
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Prof. Tao Wang
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